

“Employee Training Management System”

A PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER ENGINEERING

**SARDAR PATEL COLLEGE OF ENGINEERING,
BAKROL**



Gujarat Technological University, Ahmedabad

[April 2025]



**Sardar Patel College of Engineering,
Bakrol**

CERTIFICATE

This is to certify that the project report submitted along with the project entitled “**Employee Training Management System**” has been carried out by **Shah Deep Mukeshkumar** under my guidance in partial fulfilment for the degree of Bachelor of Engineering in **COMPUTER ENGINEERING**, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2025.

Mr. Jigar Dalwadi

Internal Guide

Mr. Jigar Dalwadi

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GUJARAT TECHNOLOGICAL UNIVERSITY

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Internship Project Report	Completed
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Name of Guide : Mr. Jigar Ashwinbhai Dalvadi

Signature of Student : _____

*Signature of Guide : _____

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HR/TEPL/193

01.01.2025

To,
Sardar Patel College of Engineering,
Bakrol.

Permission for Internship

Dear Sir/M'am,

With reference to your letter dated.01.01.2025, we are pleased to grant permission to **Mr.Rudresh K Panchal/Mr.Deep M Shah & Mr.Harsh N Jain** student of **Computer Engineering for Internship** in our organization from **20.01.2025 to 19.04.2025**.

The following points are to be adhered by the student.

- Laptops, Pen-Drives & other hard drives are not permitted inside the company premises.
- Appropriate dressing & grooming will be appreciated. Jeans & T Shirts are not allowed.
- Female students have to wear dress and Kurtis only.
- Photography & Videography is strictly prohibited in the premises.
- Trainee/s has to strictly comply with the Company's timings & schedule.
- In case of any casualty the Company will not be held responsible.
- On Completion of training the trainee/s has to submit a copy of Training Report duly signed by concern guide to obtain the Training Certificate.
- In case of any ambiguity / difficulty, the trainee is required to approach HR Department. The detailed guideline are attached overleaf.

For, Tech Elecon Pvt.Ltd.



Nirali Trivedi
Group HR Head
Cc. Satyam Raval

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**Sardar Patel College of Engineering,
Bakrol**

DECLARATION

We hereby declare that the Internship / Project report submitted along with the Internship / Project entitled “**Employee Training Management System**” submitted in partial fulfilment for the degree of **Bachelor of Engineering in Computer Engineering** to **Gujarat Technological University, Ahmedabad**, is a Bonafide record of original project work carried out by me at **SARDAR PATEL COLLEGE OF Engineering, BAKROL** under the supervision of Mr. Jigar Dalvadi and that no part of this report has been directly copied from any students’ reports or taken from any other source, without providing due reference.

Name of the Student

Shah Deep Mukeshkumar

Sign of Student

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to all those who have contributed to the successful completion of this project.

First and foremost, I extend my sincere appreciation to my project guide, **Mr. Satyam Raval**, for his invaluable guidance, continuous support, and insightful feedback throughout this project. His expertise and encouragement have been instrumental in shaping this work and overcoming challenges along the way.

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I would also like to express my profound appreciation to my family, especially my parents, for their unwavering support, patience, and encouragement throughout this journey. Their belief in me has been a constant source of strength and motivation.

Lastly, I acknowledge the contributions of everyone who, directly or indirectly, played a role in the completion of this project. Your time, knowledge, and encouragement are deeply appreciated.

Thank you all.

ABSTRACT

The Employee Training Management System (ETMS) is an advanced web and mobile platform that automates and streamlines employee training, assessments, and performance tracking. It allows employees to access training materials, complete assessments, and monitor progress in real-time. Trainers and admins can manage resources, assign projects, and review performance via dashboards. The system leverages automated evaluations, personalized recommendations, and robust analytics to enhance skill development. It ensures security through role-based access and encrypted data, while notifications keep users updated on tasks. ETMS ultimately promotes continuous learning and professional growth in organizations.

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Abbreviations

.NET	Dot Net (Microsoft framework)
AI	Artificial Intelligence
API	Application Programming Interface
ASP.NET	Active Server Pages .NET
C#	C Sharp
CDN	Content Delivery Network
CE-I	Continuous Evaluation I
CE-II	Continuous Evaluation II
CSS	Cascading Style Sheets
DBA	Database Administrator
DevOps	Development and Operations
DFD	Data Flow Diagram
ETMS	Employee Training Management System
HTML	Hypertext Markup Language
IDE	Integrated Development Environment
IIS	Internet Information Services
IT	Information Technology
LMS	Learning Management System
MVC	Model-View-Controller
ORM	Object-Relational Mapping
PDF	Portable Document Format
QA	Quality Assurance
RDBMS	Relational Database Management System
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
SSMS	SQL Server Management Studio
UI	User Interface
UAT	User Acceptance Testing
UX	User Experience

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CHAPTER 1

1.0 OVERVIEW OF THE COMPANY

1.1 HISTORY:

Tech Elecon Pvt. Ltd. is the It division of the Elecon group of companies and has more than 25 years of experience in the fields of hardware, software, and networking solutions.

It is situated in the heart of Vithal Udyognagar Industrial Estate and in the proximity of the educational town of Vallabh Vidyanagar. Tech Elecon all set to reach new heights in the field of IT solutions.

Tech Elecon is ready with all sorts of solutions and delivers any application that is web based and further our solutions are designed to adapt your business rather than your business adapting the software. Their solutions are 100% fruitful and empower you to take control of client's business online and in real time.

Tech Elecon have more than 100 employees with specialized skills in software development, custom software development, and e-commerce software development using custom software programming including .NET, C#.NET, PHP, and Open Source and Oracle. Tech Elecon delivers quality products and services with a focus on integrating the same with existing technologies, providing the required automation to our customers to help them achieve their business objectives.

Mr. Nilesh Naik, the company's Vice President, is at the helm of the Tech Elecon organization. Mr. Satyam Raval, as Deputy General Manager, and after that, Manager and Associate Manager positions are listed. At the bottom, there are trainees at entry level, who follow up to engineer, senior engineer, also executive and senior executive manager.

1.2DIFFERENT PRODUCT / SCOPE OF WORK:

Tech Elecon has extensive experience in providing IT services and has successfully adapted to technological advancements, making it the leading IT infrastructure management service provider in the region. Our cutting-edge delivery model covers all the stages of the solution lifecycle, including planning, deploying, managing, maintaining, auditing, upgrading, and improving.

Tech Elecon recognize that each client has unique needs and expectations when it comes to

infrastructure and service providers. Our clients have the flexibility to choose from a wide range of IT infrastructure management and performance services based on their specific requirements. They can opt for on-site services on hybrid solutions that include on site troubleshooting and support services.

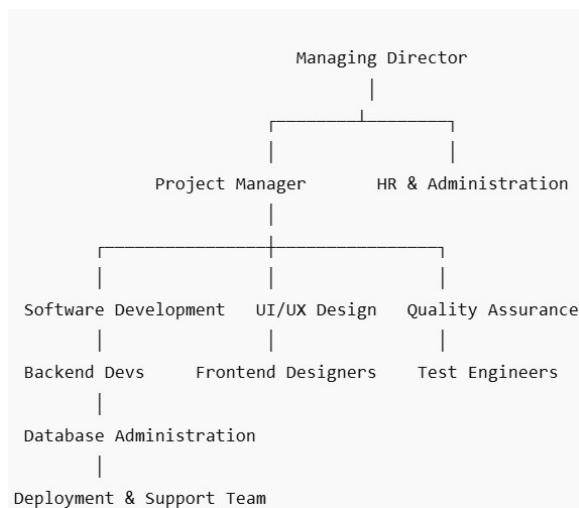
Tech Elecon provides various services for business:

- Hardware maintenance and repairing
- Service desk management
- Desktop management
- Network management
- Messaging administrator
- Back-up management

Other services:

- Software Development Services
- Software Licensing
- Microsoft Product Implementation
- Linux Servers/ Desktop Implementation

1.3 Organization chart



1.4 Capacity of plant

Although Tech Elecon Pvt. Ltd. operates in the **IT and software development sector**, the concept of "plant capacity" is interpreted in terms of **technical infrastructure, project handling capability, and team productivity**.

Technical Capacity Highlights:

- **Employee Strength:** 50+ skilled professionals across development, testing, design, and management.
- **Workstations:** Fully equipped development labs with modern PCs and high-speed internet connectivity.
- **Project Handling Capability:** Capable of managing 5–8 mid-to-large scale projects simultaneously.
- **Cloud Infrastructure:** Uses cloud-based services like **Microsoft Azure** for hosting and deployment, ensuring scalability and uptime.
- **Version Control & CI/CD:** Integrated **GitHub** and **Azure DevOps** pipelines for continuous integration and delivery.

The infrastructure and human resource capacity of Tech Elecon Pvt. Ltd. enables the company to undertake comprehensive and complex software development projects, such as the ETMS, with efficiency and reliability.

Chapter 2

2.0 Overview of Different Departments and Process Layout

Tech Elecon Pvt. Ltd. is structured into several functional departments that work collaboratively to ensure seamless project development and delivery. The company follows a modern and systematic layout in handling its software development lifecycle, ensuring effective communication between departments and efficient execution of processes.

The primary departments involved in the development and deployment of software products like the **Employee Training Management System (ETMS)** include:

- **Project Management Department**
- **Software Development Department**
- **Quality Assurance (QA) Department**
- **UI/UX Design Department**
- **Database Administration (DBA) Department**
- **Deployment and Support Department**

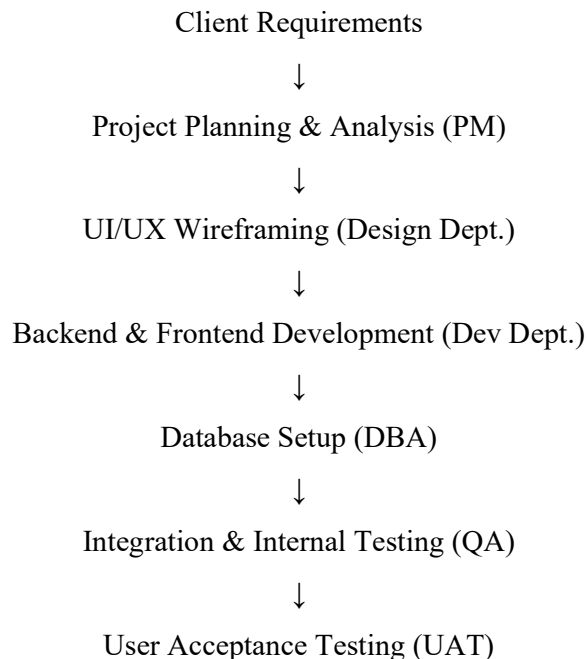
2.1 Department-wise Work Details

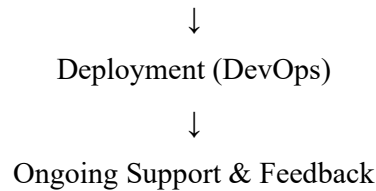
- **Project Management Department**
Responsible for project planning, requirement gathering, task allocation, and communication between client and technical teams.
- **Software Development Department**
Focuses on writing code, implementing features using ASP.NET Core MVC, Entity Framework Core, and integrating with SQL Server.
- **Quality Assurance Department**
Handles testing of the application through manual and automated techniques to ensure bugs are eliminated and the system works as expected.
- **UI/UX Design Department**
Designs user-friendly interfaces ensuring good user experience using responsive design techniques and frontend technologies.
- **Database Administration Department**
Manages creation, optimization, and maintenance of SQL Server databases and ensures secure data handling.

2.2 Technical Specifications of Major Tools and Technologies Used

Department	Technology / Tool Used	Specification
Software Development	ASP.NET Core MVC	Backend framework for building web applications
	Entity Framework Core	ORM for database interactions
	Visual Studio 2022	IDE for development
UI/UX Design	HTML, CSS, JavaScript, Bootstrap	Frontend technologies for responsive design
Database Administration	Microsoft SQL Server	RDBMS used for storing training, user, and test data
Project & Task Management	Azure DevOps / Trello / Jira	Tools for task allocation and sprint management
Quality Assurance	Selenium / Manual Testing Tools	For automated and manual testing of features
Deployment and Support	IIS / GitHub / Azure	Deployment and version control

2.3 Schematic Layout of Development Process





2.4 Detailed Explanation of Each Stage

1. Client Requirements & Planning

Initial meetings are held with stakeholders to define the scope of ETMS. Functional and non-functional requirements are gathered and documented.

2. Design & Prototyping

UI/UX designers prepare mockups and interactive prototypes. The goal is to ensure usability for Admin, Trainer, and Employee roles.

3. Development

The system is developed in modules—user authentication, test management, report generation, learning material management, etc. Features are implemented using ASP.NET Core MVC and Entity Framework.

4. Database Integration

Database models and relationships are created and mapped using EF Core. Key tables include Users, Tests, Answers, Projects, and Reports.

5. Testing

Rigorous testing is conducted to verify the correctness and security of the system. Bugs are tracked and resolved before final deployment.

6. Deployment

Once approved, the application is deployed on the company server. Post-deployment, minor improvements and monitoring continue based on real-world usage.

7. Maintenance & Support

The support team monitors the application, handles bug reports, and ensures smooth operation for end-users.

CHAPTER 3

3.0 INTRODUCTION TO PROJECT

3.1 INTERNSHIP SUMMARY:

Training and skill development are critical components of an organization's success, yet many companies face challenges in efficiently managing employee training programs, tracking assessments, and evaluating performance. During my industrial internship, I worked on developing an Employee Training Management System (ETMS), a web-based application designed to streamline employee training, assessments, and progress tracking.

The primary objective of this project was to create a centralized platform that allows employees to take assessments, receive training, and track their performance, while enabling admins and trainers to manage training materials, assign projects, and generate reports. By implementing automated assessment evaluations, training progress tracking, and reporting mechanisms, the system helps improve employee skill development and organizational efficiency.

The project involved designing a secure, scalable, and role-based access system using ASP.NET Core MVC, Entity Framework Core, and SQL Server. The system provides video lectures, interactive training modules, and real-time feedback, ensuring a structured learning experience for employees. The ultimate goal was to enhance the training process through automation, analytics, and an intuitive interface.

3.2 PURPOSE:

The purpose of the Employee Training Management System (ETMS) is to provide a structured and automated approach to employee training and development. Traditional training methods involve manual tracking, scattered learning materials, and inefficient performance evaluation, leading to inconsistent learning outcomes and increased administrative effort.

This project aims to:

- Centralize employee training management by providing a single platform for assessments, learning materials, and progress tracking.
- Enhance training efficiency with automated test evaluations.

- Improve performance evaluation by tracking employee progress, quiz results, and feedback mechanisms.
- Streamline administration by enabling trainers and admins to assign projects, monitor training and generate reports.
- Automate notifications and reminders for training schedules, test deadlines.

By achieving these objectives, ETMS ensures that organizations optimize employee learning, enhance workforce skills, and maintain a well-structured training process.

3.3 OBJECTIVE:

The Employee Training Management System (ETMS) is developed to provide an efficient and automated solution for managing employee training. The key objectives of this project are:

- **User Authentication & Role Management:** Implement secure login mechanisms and role-based access control (RBAC) for employees, trainers, and admins.
- **Assessment Management:** Develop an automated system for conducting **aptitude** and technical tests, along with automatic result evaluation.
- **Training Module:** Enable trainers to upload and manage video lectures, PDFs, and interactive training content, ensuring a structured learning experience.
- **Project Tracking:** Allow trainers to assign projects and monitor project submissions, while employees can submit project and receive feedback.
- **Performance Evaluation:** Track employee progress through quizzes, project submissions, and feedback mechanisms.
- **Automated Notifications:** Implement an automated alert system for training deadlines and test schedules.

By fulfilling these objectives, ETMS provides a scalable, user-friendly, and efficient approach to employee training and skill development.

3.4 SCOPE:

The Employee Training Management System (ETMS) is designed to support organizations, HR teams, and training managers in efficiently managing employee training programs. The system covers the following areas:

Training & Assessment Management

- Conduct aptitude and technical tests for employees.
- Provide automated result evaluation.
- Upload and manage video lectures, documents, and interactive training materials.

Employee & Admin Dashboards

- Employees can view training schedules, track progress, and submit project.
- Trainers can monitor employee performance, upload new learning content, and review assessments.
- Admins can manage users and control role-based access and generate reports.

Performance Evaluation & Reporting

- Generate performance reports based on quizzes, training completion, and project evaluations.
- Provide feedback mechanisms for continuous improvement.
- Offer training completion certificates for employees who meet learning objectives.

Security & Authentication

- Implement secure login and access management using ASP.NET Core Identity Framework.
- Ensure data security for employee records and training materials.

By covering these aspects, ETMS ensures that organizations have a comprehensive and structured solution for employee training management.

3.5 TECHNOLOGY AND LITERATURE REVIEW:

3.5.1 Technology Used:

- **Backend:** ASP.NET Core MVC, Entity Framework Core
- **Frontend:** Bootstrap, JavaScript, jQuery
- **Database:** SQL Server
- **Authentication:** ASP.NET Core Identity Framework
- **Storage:** Local file storage for training materials (videos, PDFs)

3.5.2 Literature Review:

Research in the field of employee training management has explored various solutions, methodologies, and best practices aimed at improving training effectiveness and performance tracking. Several past solutions have attempted to address similar challenges through different approaches:

Existing Learning Management Systems (LMS)

1. **Moodle & Blackboard:** Traditional LMS platforms like Moodle and Blackboard provide structured learning environments for corporate training. However, they primarily focus on content delivery rather than real-time performance tracking and automated training recommendations.
2. **SAP SuccessFactors:** A corporate training platform that integrates HR processes with learning management but lacks detailed personalized assessment-based training recommendations like ETMS.
3. **Google Classroom & Coursera for Business:** These platforms offer cloud-based solutions for corporate learning but are not tailored for customized organizational training workflows and real-time progress tracking.

Past Research & Studies

1. **Personalized Learning Algorithms:** Studies in adaptive learning (Brown & Smith, 2021) highlight the importance of AI-driven systems that adapt training content based on user performance. The ETMS integrates this by analyzing employee test results and suggesting tailored training paths.
2. **Gamification in Employee Training:** Research by Lee et al. (2020) discusses the effectiveness of interactive and gamified learning experiences in increasing employee engagement. Our ETMS incorporates interactive quizzes, feedback loops, and structured assessments.
3. **Automated Training Recommendations:** A study by Nguyen & Patel (2019) explored how machine learning models can analyze assessment data to provide intelligent training suggestions. ETMS applies this approach to automate skill gap analysis and recommend learning modules based on performance.

3.6 INTERNSHIP PLANNING:

3.6.1 Internship Development Approach and Justification

The development of ETMS followed an iterative and modular approach, ensuring flexibility and incremental improvements. This approach was selected due to its effectiveness in handling complex, multi-functional systems like employee training platforms.

3.6.2 Internship Effort and Time, Cost Estimation

The estimated effort and cost were determined based on the scope, required technology stack, and available resources. The project leveraged open-source frameworks (ASP.NET Core, Bootstrap, SQL Server Express) to minimize costs, while cloud storage or local hosting solutions were considered for training material management.

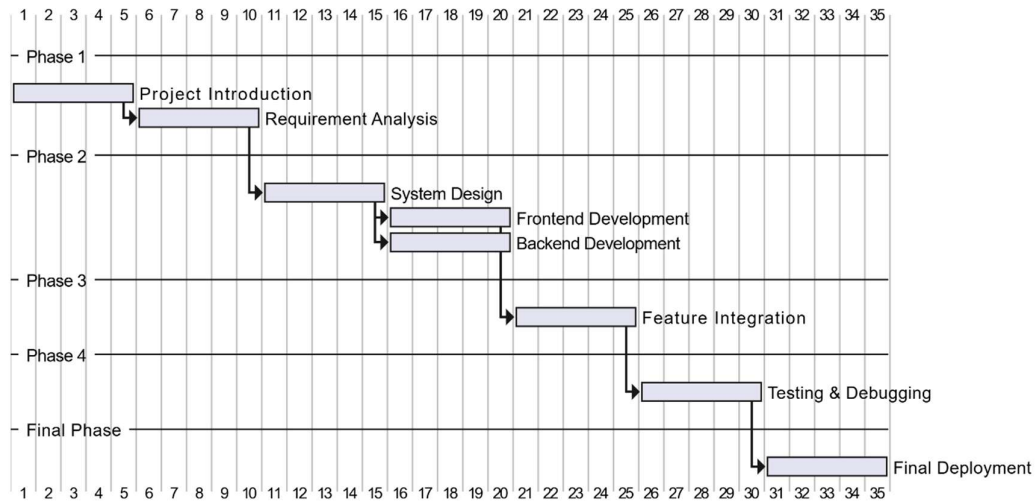
3.6.3 Roles and Responsibilities

- **Project Developer (Intern):** Responsible for coding, database management, UI development, and system integration.
- **Project Mentor:** Provided technical guidance, code reviews, and overall project supervision.
- **Peers & Reviewers:** Assisted in testing, debugging, and providing feedback for improvement.

3.6.4 Group Dependencies

Even though this project was developed independently, there were certain dependencies on external sources. The project relied on external APIs for authentication, cloud storage solutions for training materials, and feedback from trainers and HR professionals to refine system features. Collaboration with industry experts and academic mentors ensured that the project aligned with real-world training management needs.

3.7 Internship Scheduling (Gantt Chart)



CHAPTER 4

4.0 SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM:

Currently, most organizations rely on manual processes, spreadsheets, or outdated Learning Management Systems (LMS) to manage employee training. These systems require significant administrative effort, lack real-time tracking, and often fail to provide personalized training recommendations.

Additionally, many training platforms focus primarily on content delivery rather than interactive assessments, feedback, and performance tracking. This results in inefficiencies, reduced engagement, and difficulty in measuring employee growth over time.

4.2 PROBLEMS & WEAKNESS OF CURRENT SYSTEM:

The existing training management systems present several challenges:

- **Manual Training Management** – HR teams struggle with tracking employee progress, training completion, and assessment results manually.
- **Lack of Real-Time Feedback** – Employees do not receive instant feedback on assessments, making it difficult to track learning progress.
- **No Automated Recommendations** – Current systems do not suggest personalized learning paths based on assessment results.
- **Difficulty in Performance Evaluation** – There is no structured evaluation process for tracking employee performance over time.
- **Limited Reporting Capabilities** – Most training systems lack comprehensive analytics to measure training effectiveness.
- **No Automated Notifications** – Employees often miss training deadlines due to the absence of reminders and alerts.

4.3 REQUIREMENTS OF SYSTEM:

The Employee Training Management System (ETMS) aims to overcome these challenges by implementing the following requirements:

Functional Requirements:

- **User Authentication & Role-Based Access** – Employees, Trainers, and Admins have different access levels.
- **Automated Assessments & Evaluations** – System conducts aptitude and technical tests with automatic scoring.
- **Training Content Management** – Trainers can upload videos, PDFs, and interactive content.
- **Project Tracking** – Employees receive projects with deadlines, and trainers can provide feedback.
- **Performance Monitoring & Feedback** – Real-time tracking of employee progress and evaluations.
- **Automated Notifications** – Alerts for test deadlines and performance updates.
- **Reporting & Analytics** – Generate training progress reports, performance insights, and skill evaluations.

Non-Functional Requirements:

- **Security & Authentication** – Use ASP.NET Core Identity Framework for secure logins.
- **Scalability** – Handle multiple users and large training data efficiently.
- **User-Friendly Interface** – Ensure an intuitive UI for employees, trainers, and admins.
- **Performance Optimization** – Enable fast data retrieval, report generation, and content delivery.

4.4 SYSTEM FEASIBILITY:**4.4.1 Contribution to Organizational Objectives:**

The ETMS contributes to enhancing workforce skills, streamlining training processes, and improving employee engagement. It ensures structured learning, automated assessments, and effective performance tracking.

4.4.2 Implementation Using Current Technology & Budget:

- The system utilizes cost-effective and scalable technologies like ASP.NET Core, SQL Server, and Bootstrap.
- Open-source frameworks reduce development costs.
- The project follows an Agile methodology, ensuring delivery within the allocated timeframe and resources.

4.4.3 Integration with Existing Systems:

ETMS can be integrated with HR systems, third-party APIs, and existing training platforms for seamless data exchange. The system also supports data import/export features.

4.5 PROCESS IN NEW SYSTEM:

Step 1: User Authentication & Role-Based Access

- Employees, Trainers, and Admins log in with secure credentials.
- Access permissions are based on roles.

Step 2: Assessment & Training Module

- Employees take automated aptitude and technical tests.
- Trainers upload training videos, PDFs, and assignments.

Step 3: Assignment & Project Tracking

- Employees receive project assignment with deadlines.
- Trainers evaluate submissions and provide feedback.

Step 4: Performance Review & Analytics

- System tracks quiz scores, project completion, and training progress.
- Reports are generated for employee growth analysis.

Step 5: Notifications & Reminders

- Employees receive email/SMS alerts for upcoming training tests and deadlines.

4.6 Features of New System

The newly developed **Employee Training Management System (ETMS)** offers a robust and user-friendly platform to streamline employee training, project tracking, and performance evaluation. The key features of the system are as follows:

1. **Role-Based Access Control**

The system supports different roles such as Admin, Trainer, and Employee, each with tailored access and functionalities.

2. **Employee Registration and Management**

Admins can register, edit, view, and delete employee details efficiently.

3. **Trainer-Employee Assignment**

Admins can assign employees to trainers based on their training modules or departments.

4. **Test Management System**

Trainers or Admins can create and assign tests to employees with time constraints, auto-submission on timeout, and one-time access control.

5. **Answer Sheet & Result View**

Employees can view their submitted answers and performance results after test completion.

6. **Learning Material Management**

Trainers can upload PDFs, videos, and presentations related to the training modules, which employees can access based on their assignment.

7. **Project Assignment & Submission**

Trainers can create projects with descriptions and deadlines, assign them to employees, and review submissions uploaded as ZIP files.

8. **Employee Performance Report**

Trainers/Admins can generate comprehensive performance reports that include test marks, project marks, and feedback questions.

9. **Dashboard for Each Role**

All roles have a dedicated dashboard summarizing key actions, recent activities, and quick navigation to core features.

10. Secure Authentication

The system uses ASP.NET Identity for secure login with hashed passwords and email-based authentication.

11. User-Friendly Interface

The UI is designed to be intuitive and responsive for a smooth user experience across devices.

12. Real-Time Alerts & Notifications

Users are notified of project assignments, deadlines, test schedules, and system updates in real-time.

4.7 LIST OF MAIN MODULES:

- **User Authentication & Role Management** – Secure login & role-based access.
- **Assessment Management** – Conduct automated tests & evaluations.
- **Training Module** – Manage training materials (videos, PDFs, interactive content).
- **Project Tracking** – Assign and track employee tasks.
- **Performance Monitoring** – Monitor training completion & feedback.

4.8 Selection of Software

The selection of appropriate software tools and technologies plays a crucial role in the successful development and deployment of the **Employee Training Management System (ETMS)**. After careful analysis of project requirements, scalability, performance, and developer familiarity, the following software stack was selected:

1. Frontend Development

- **Language:** HTML, CSS, JavaScript
- **Framework:** Bootstrap (for responsive design)
- **Razor View Engine:** Used within ASP.NET Core MVC for dynamic content rendering

2. Backend Development

- **Technology:** ASP.NET Core MVC
- **Language:** C#

- **Framework Version:** .NET 6.0
- ASP.NET Core was chosen due to its robustness, high performance, built-in security, and MVC architecture support.

3. Database

- **Database Management System:** Microsoft SQL Server
- **ORM Tool:** Entity Framework Core
- Chosen for its seamless integration with .NET, strong relational support, and ease of migrations and LINQ querying.

4. Authentication & Security

- **ASP.NET Identity:** Provides secure user authentication, role-based access control, and password hashing.

5. IDE & Tools

- **Visual Studio 2022:** For efficient development with IntelliSense, debugging, and integrated Git support.
- **SQL Server Management Studio (SSMS):** For database design, query execution, and monitoring.
- **Postman:** For testing backend APIs (if applicable).
- **GitHub:** Version control and collaboration.

6. Deployment

- **IIS (Internet Information Services):** For local or production server deployment.
- **.NET CLI Tools:** For building, publishing, and managing the project.

CHAPTER 5

5.0 SYSTEM DESIGN

5.1 SYSTEM DESIGN AND METHDOLOGY:

Development Approach:

The Employee Training Management System (ETMS) follows an Agile development methodology, ensuring iterative improvements with regular feedback from mentors and industry experts. Each sprint focused on delivering specific functionalities, such as user authentication, learning material management, and assessment modules.

System Architecture:

The ETMS is built on a client-server model using a three-tier architecture:

- **Presentation Layer:** Developed using ASP.NET Core MVC, Bootstrap, and jQuery, providing a responsive and user-friendly interface.
- **Business Logic Layer:** Implemented in .NET Core, handling user authentication, learning material management, assessments, and reporting.
- **Data Layer:** Managed by SQL Server, storing user data, training materials, assessments, and performance reports.

Modular Design: The system is divided into independent modules (e.g., Trainer Dashboard, Admin Dashboard, Employee Dashboard), allowing for easy updates and maintenance.

5.2 INPUT/OUTPUT INTERFACE DESIGN:

User Interface (UI) Design:

The ETMS features role-specific dashboards for Admins, Trainers, and Employees, ensuring a seamless user experience:

Trainer Dashboard:

- Displays assigned employees.

- Allows Trainers to assign learning materials to employees.
- Provides insights into training progress and assessment results.

Employee Dashboard:

- Shows assigned training materials and progress tracking.
- Displays notifications for newly assigned materials and assessments.
- Allows employees to take assessments and view performance reports.

Admin Dashboard:

- Manages Users, Trainers, Employees, and Learning Materials.
- Provides an overview of assessment results and employee performance.
- Enables trainer-employee assignments and overall system monitoring.

5.2.1 State and Transition of Learning Materials:

The following states represent how training materials are assigned and interacted with:

- Available: Trainers upload new learning materials.
- Assigned: Trainers assign materials to specific employees.
- Viewed: Employees access assigned materials.
- Completed: Employees finish reviewing assigned materials.

5.2.2 Access Control and Security**The ETMS implements Role-Based Access Control (RBAC):**

- Admin: Manages users, trainers, and training materials.
- Trainer: Assigns learning materials and tracks employee progress.
- Employee: Views assigned materials, completes assessments, and tracks progress.

Security Measures:

- Authentication: ASP.NET Identity ensures secure logins.
- Authorization: Role-based permissions restrict unauthorized access.
- Data Protection: SQL Server uses encrypted storage for sensitive data.

- **Input Validation:** Client-side and server-side validation to prevent data breaches.
- **Audit Logging:**
 - Every action (e.g., material upload, assessment completion) is logged for compliance and tracking.
- **Error Handling and Notifications:**
 - Implemented error messages for invalid actions.
 - Email/SMS notifications for employees upon assignment of new materials.

5.3 DATABASE DESIGN

1. AspNetUsers:

	Name	Owner	Type	Created_datetime				
1	AspNetUsers	dbo	user table	2025-02-21 10:55:47.403				
	Column_name	Type	Computed	Length	Prec	Scale	Nullable	
1	Id	varchar	no	450			no	
2	Email	varchar	no	255			no	
3	PasswordHash	text	no	16			no	
4	UserName	varchar	no	255			no	
5	Role	varchar	no	50			no	
6	NormalizedEmail	nvarchar	no	512			yes	
7	NormalizedUserName	nvarchar	no	512			yes	
8	EmailConfirmed	bit	no	1			no	

Fig 5.1 AspNetUsers Table

2. EmployeeModules Table:

	Name	Owner	Type	Created_datetime				
1	EmployeeModules	dbo	user table	2025-03-30 11:07:01.033				
	Column_name	Type	Computed	Length	Prec	Scale	Nullable	
1	Id	int	no	4	10	0	no	
2	EmployeeId	varchar	no	450			no	
3	ModuleId	int	no	4	10	0	no	

Fig 5.2 EmployeeModules

3.LearningMaterials Table:

	Name	Owner	Type	Created_datetime			
1	LearningMaterials	dbo	user table	2025-03-22 11:27:08.680			

	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	Id	int	no	4	10	0	no
2	ModuleId	int	no	4	10	0	no
3	FileName	nvarchar	no	510			no
4	FilePath	nvarchar	no	1000			no
5	FileType	nvarchar	no	20			no
6	UploadedOn	datetime	no	8			yes

Fig 5.3 LearningMaterials

4.Questions Table:

	Name	Owner	Type	Created_datetime			
1	Questions	dbo	user table	2025-03-02 11:05:15.487			

	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	Id	int	no	4	10	0	no
2	QuestionText	nvarchar	no	1000			no
3	OptionA	nvarchar	no	510			no
4	OptionB	nvarchar	no	510			no
5	OptionC	nvarchar	no	510			no
6	OptionD	nvarchar	no	510			no
7	CorrectAnswer	nvarchar	no	20			no
8	TestType	nvarchar	no	100			no

Fig 5.4 Questions Table

5.TestModes Table:

	Name	Owner	Type	Created_datetime			
1	TestModes	dbo	user table	2025-03-03 11:12:11.220			
	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	Id	int	no	4	10	0	no
2	Mode	nvarchar	no	200			no
3	TestType	nvarchar	no	200			no

Fig 5.5 TestModesTable

6.TestModeUpdate Table:

	Name	Owner	Type	Created_datetime			
1	TestModeUpdate	dbo	user table	2025-03-03 11:51:47.040			
	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	Id	int	no	4	10	0	no
2	TestType	nvarchar	no	200			no
3	Mode	nvarchar	no	40			no

Fig 5.6 TestModeUpdate Table

7.TestSubmissions Table:

	Name	Owner	Type	Created_datetime			
1	TestSubmissions	dbo	user table	2025-03-20 10:46:57.543			
	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	Id	int	no	4	10	0	no
2	UserId	varchar	no	450			no
3	TestType	nvarchar	no	510			no
4	SubmittedAt	datetime	no	8			yes

Fig 5.7 TestSubmissions Table

8.TrainingModules Table:

	Name	Owner	Type	Created_datetime			
1	TrainingModules	dbo	user table	2025-03-22 11:27:08.657			
	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	Id	int	no	4	10	0	no
2	TrainerId	varchar	no	450			no
3	SubjectName	nvarchar	no	510			no
4	CreatedOn	datetime	no	8			yes

Fig 5.8 TrainingModules Table

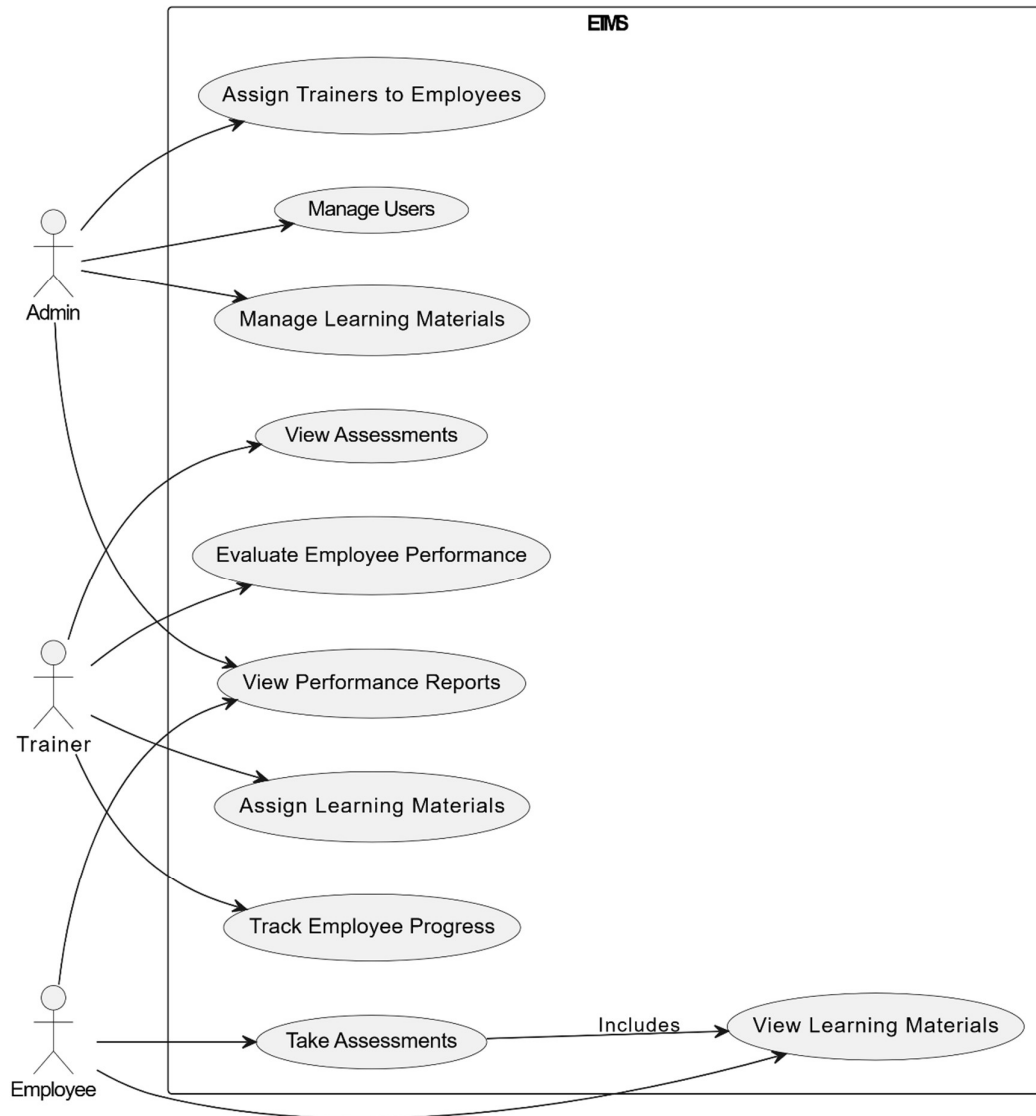
9.UserResponses Table:

	Name	Owner	Type	Created_datetime			
1	UserResponses	dbo	user table	2025-03-03 14:53:37.823			
	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	Id	int	no	4	10	0	no
2	QuestionId	int	no	4	10	0	no
3	UserId	nvarchar	no	900			no
4	SelectedOption	nvarchar	no	20			no
5	SubmittedAt	datetime	no	8			no
6	TestType	nvarchar	no	100			no

Fig 5.9 UserResponses Table

5.4 Diagrams

5.4.1 Use Case Diagram

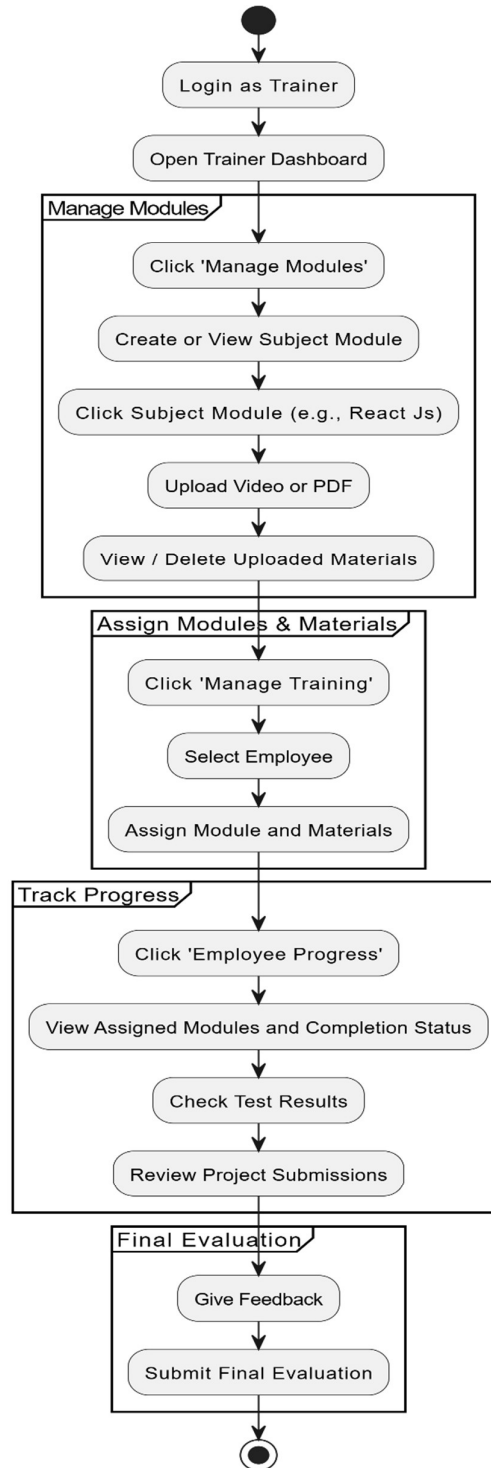


5.10 Use Case Diagram

5.4.2 Activity Diagram

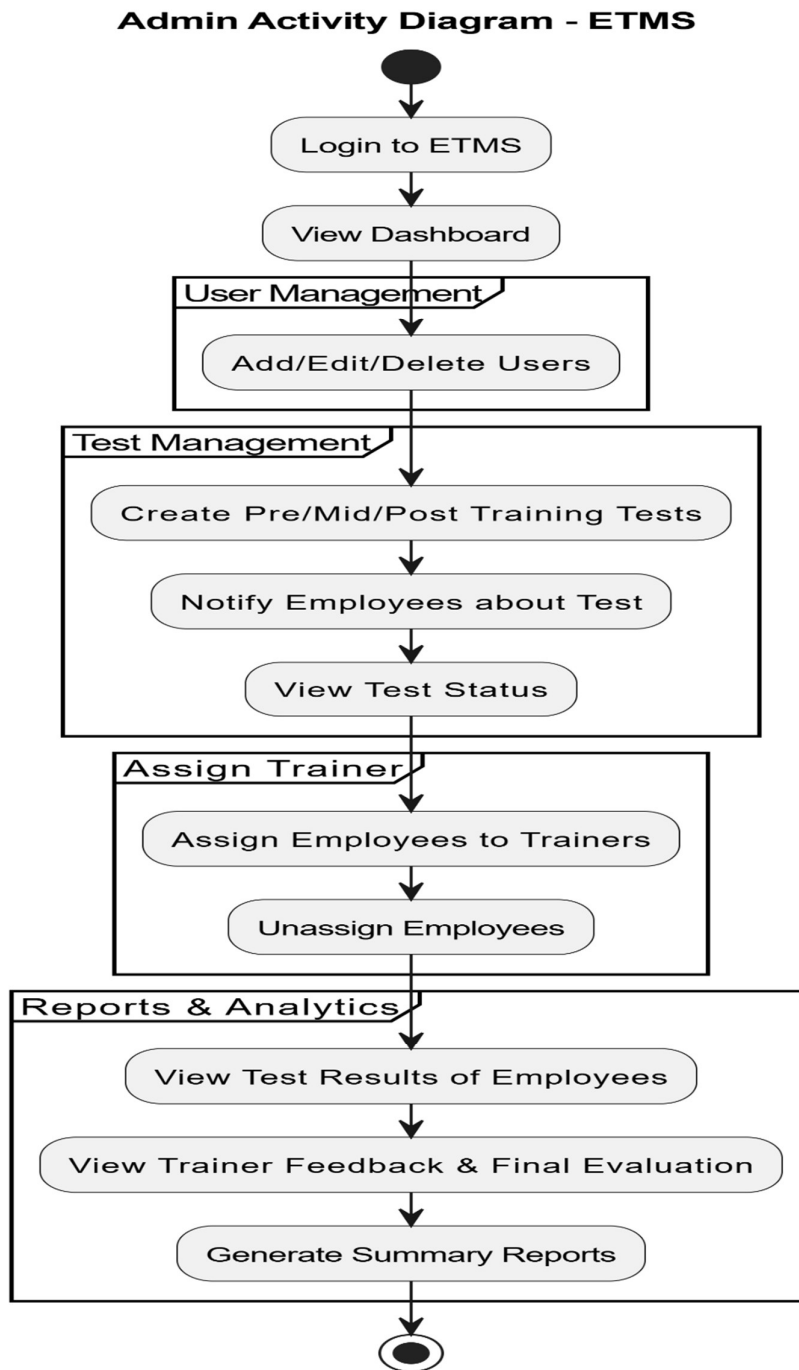
5.4.2.1 Activity Diagram (Trainer)

Activity Diagram - Trainer Dashboard Flow with Material Management and Evaluation



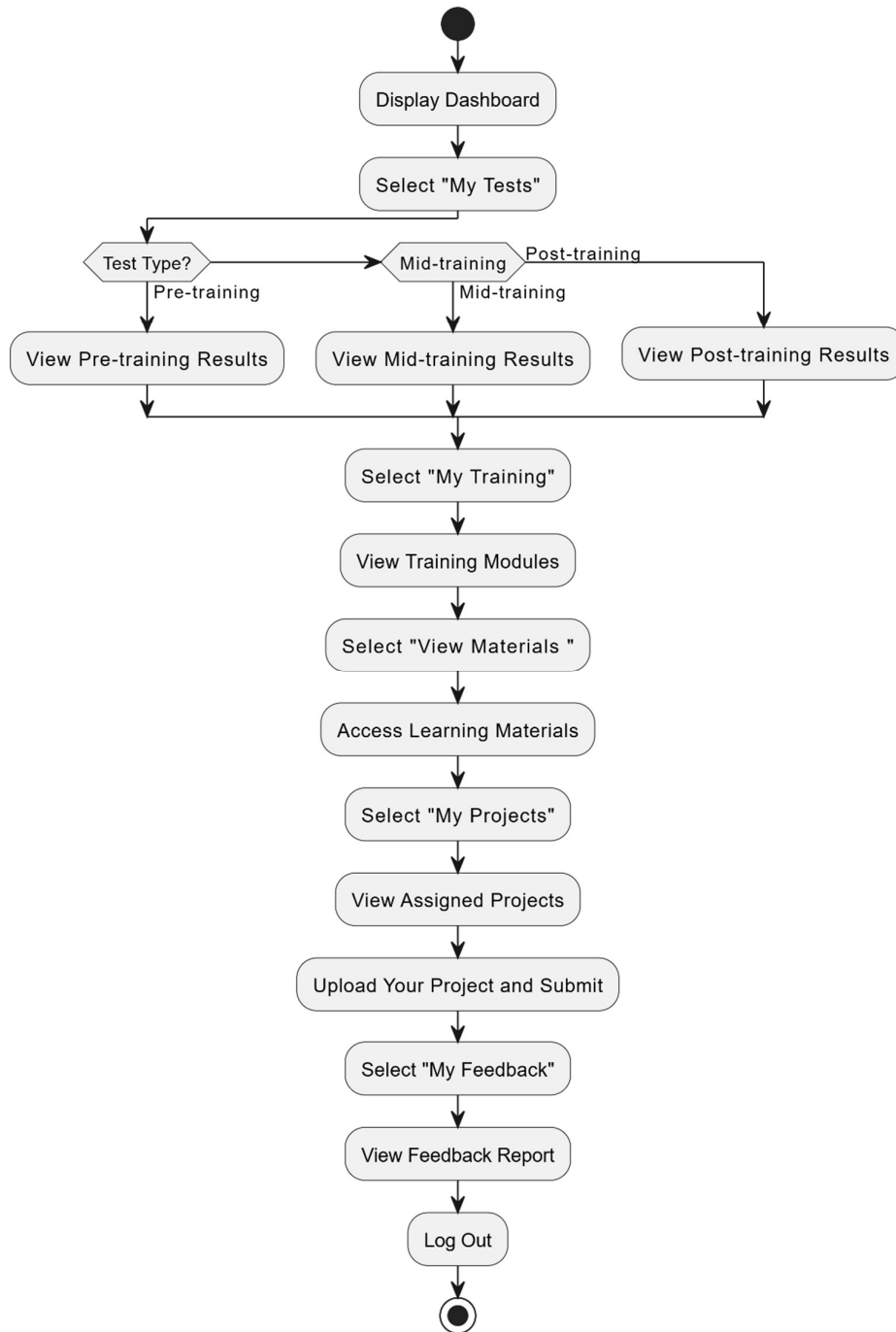
5.11 Activity Diagram (Trainer)

5.4.2.2 Activity Diagram (Admin)



5.12 Activity Diagram (Admin)

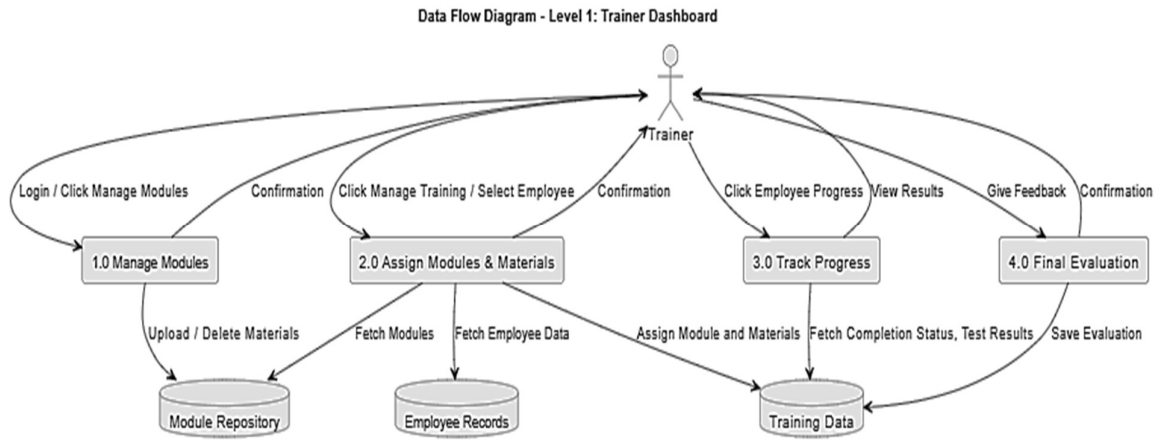
5.4.2.3 Activity Diagram (Employee)



5.12 Activity Diagram (Employee)

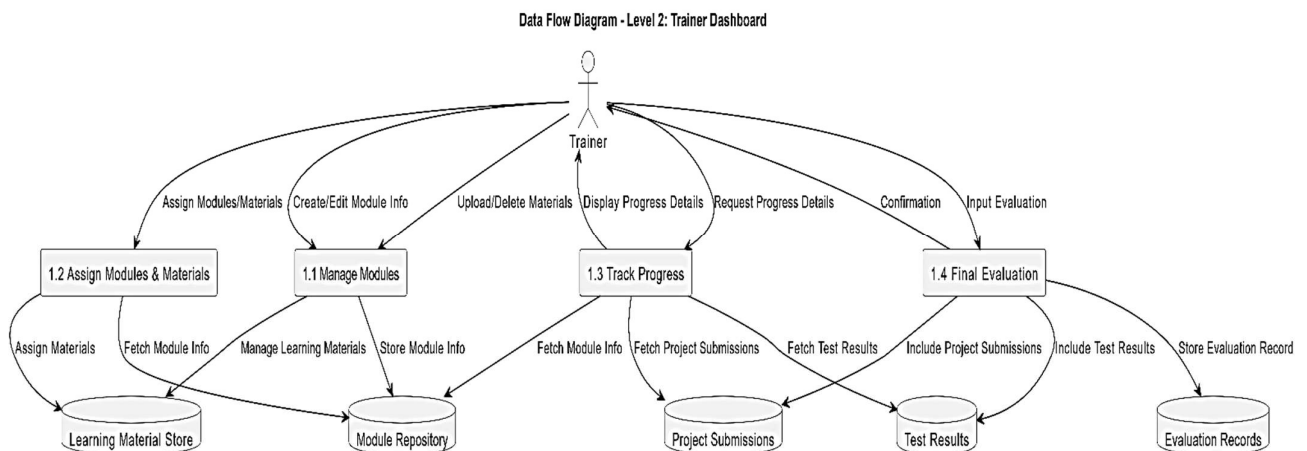
5.4.3 Data Flow Diagram

5.4.3.1 Data Flow Diagram Level 1(Trainer)



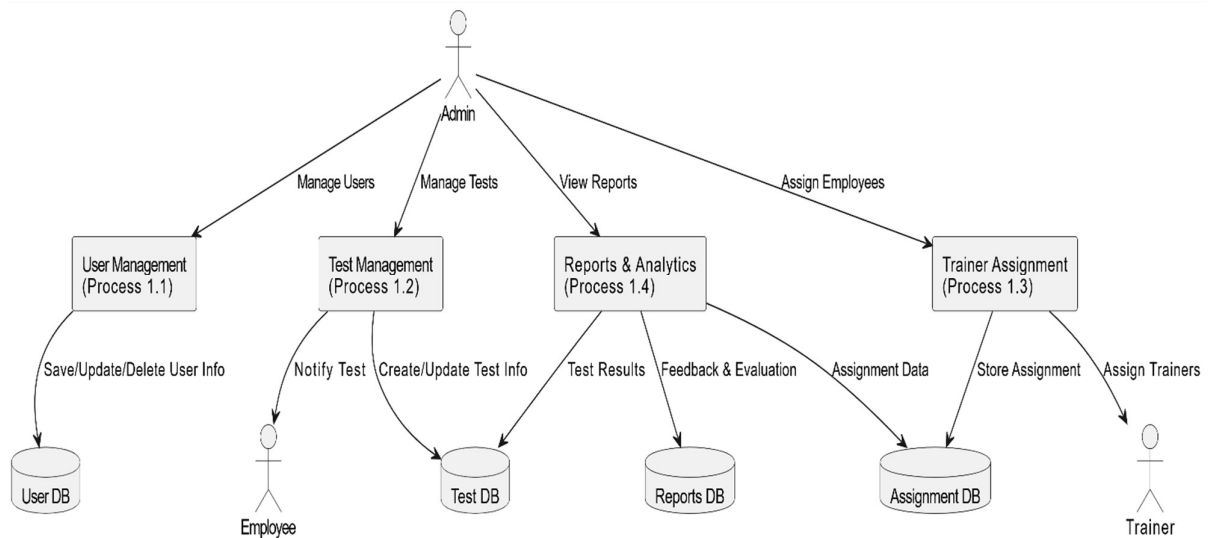
5.14 Data Flow Diagram Level 1 (Trainer)

5.4.3.2 Data Flow Diagram Level 2(Trainer)



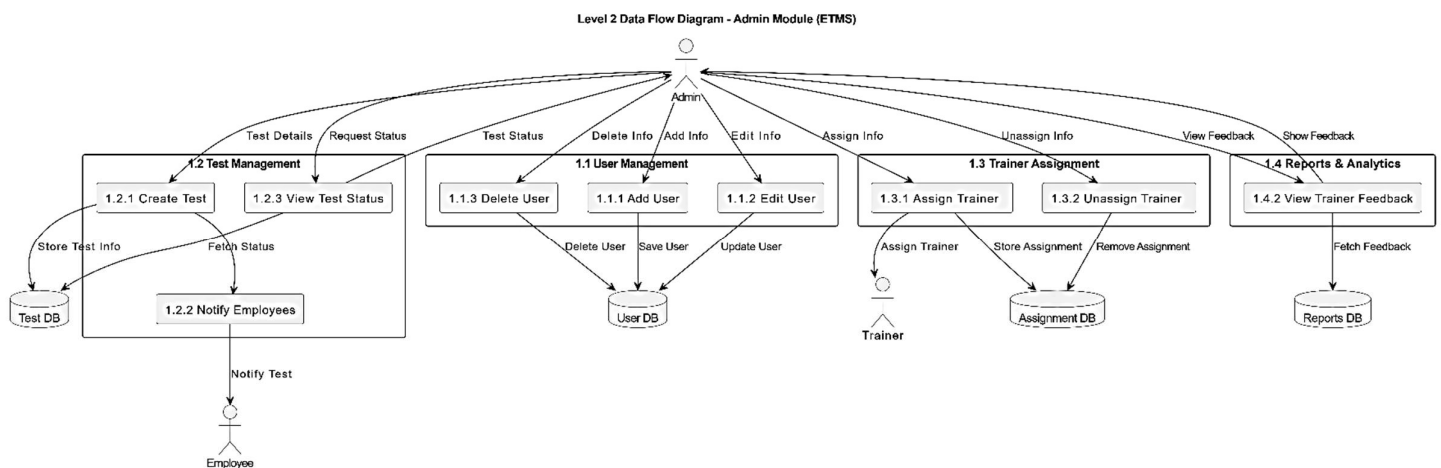
5.15 Data Flow Diagram Level 2 (Trainer)

5.4.3.3 Data Flow Diagram Level 1(Admin)



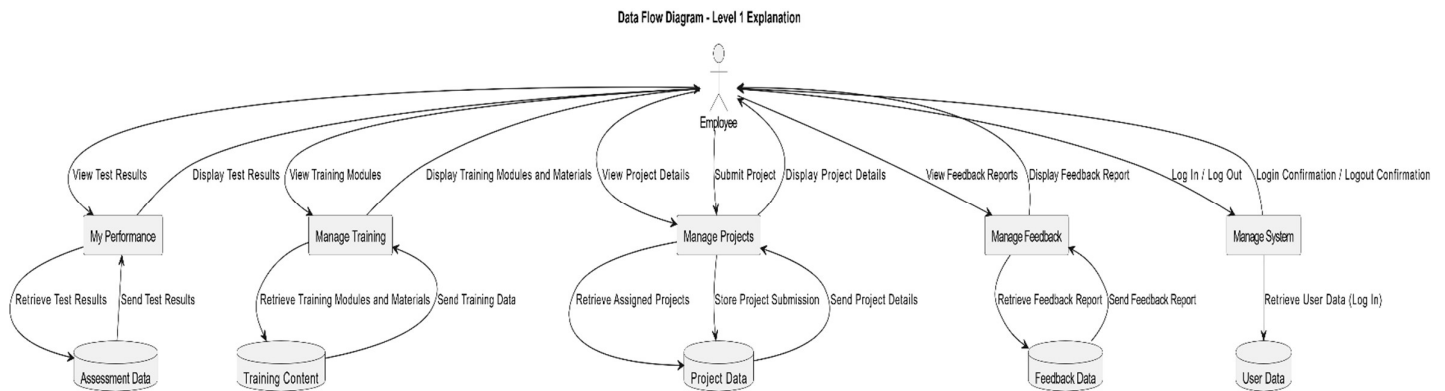
5.16 Data Flow Diagram Level 1 (Admin)

5.4.3.4 Data Flow Diagram Level 2(Admin)



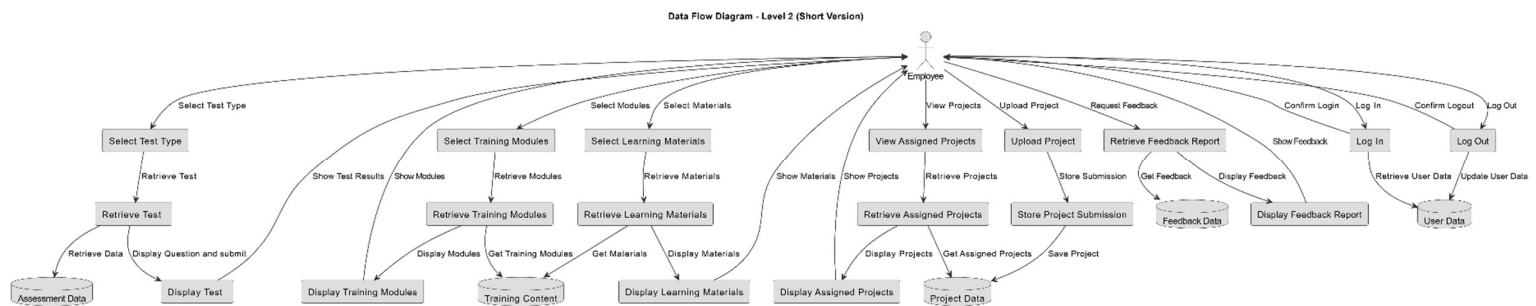
5.17 Data Flow Diagram Level 2 (Admin)

5.4.3.5 Data Flow Diagram Level 1(Employee)



5.18 Data Flow Diagram Level 1 (Employee)

5.4.3.6 Data Flow Diagram Level 2(Employee)



5.19 Data Flow Diagram Level 1 (Employee)

CHAPTER 6

6.0 IMPLEMENTATION

6.1 IMPLEMENTATION PLATFORM:

Development Environment:

- **IDE:** Visual Studio 2022 was used for coding, debugging, and testing the ASP.NET application.
- **Operating System:** Development was carried out on Windows 11.
- **Version Control:** Git was used for version control, with the repository hosted on GitHub.

Server Environment:

- **Web Server:** The application was tested on a local IIS (Internet Information Services) server. For production, Microsoft Azure is recommended.
- **Database Server:** SQL Server 2019 Express Edition was used locally, with plans for cloud-hosted SQL Server in production.

Frontend Environment:

- **Browser Compatibility:** Tested on Google Chrome, Mozilla Firefox, and Microsoft Edge.
- **Libraries:** Bootstrap 5.1.3 and jQuery 3.6.0 were included via CDN for responsiveness and fast loading.

Third-Party Services:

- **PDF Generation:** The iTextSharp library was used for generating PDFs (e.g., invoices).
- **Email Service:** SMTP was integrated using System.Net.Mail for automated email notifications.

6.2 MODULE SPECIFICATIONS:

1. User Management:

- Purpose: Handles user authentication, registration, and role-based access.
- Implementation:
 - Used ASP.NET Identity for authentication and role management.
 - Created UserManagerController for user registration, login, and role assignments.
 - Views: _ViewImports.cshtml, _ViewStart.cshtml.

2. Trainer Dashboard:

- Purpose: Enables trainers to upload, assign, and manage training materials.
- Implementation:
 - Created TrainerDashboardController for CRUD operations.
 - Views: ManageTraining.cshtml, ManageModules.cshtml and ModuleMaterial.cshtml.

3. Assessments Module:

- Purpose: Enables employees to take assessments and stores their scores.
- Implementation:
 - Created EmployeeDashboardController to handle assessments and scoring.
 - Views: AnswerQuestions.cshtml, AnswerSheets.cshtml, MyTest.cshtml, TestResults.cshtml.

4. Admin Dashboard:

- Purpose: Manages users, trainers, employees, and training materials.
- Implementation:
 - Created AdminDashboardController to handle administrative functions.
 - Views: CreateQuestion.cshtml, EditQuestion.cshtml, TestManagement.cshtml, ViewQuestions.cshtml.

6.3 RESULT AND OUTCOMES:

The implementation of Employee Training Management System yielded several key findings, results, and outcomes, demonstrating its effectiveness in addressing the challenges of current training systems. Below are the detailed outcomes, supported by visual evidence of the system's functionality through screenshots of the ETMS website pages.

6.3.1 Home page

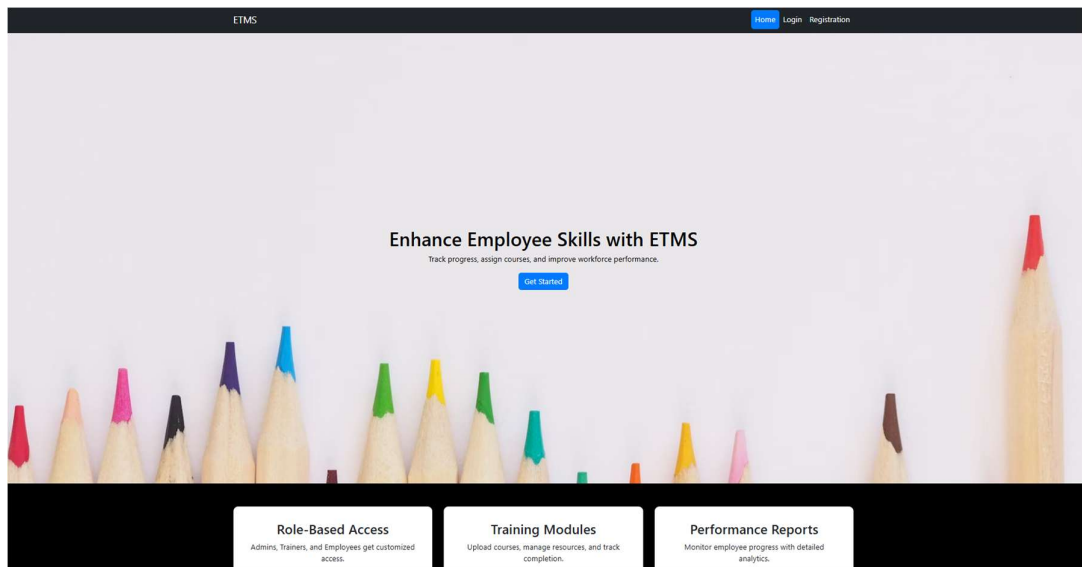


Fig 6.1 Home Page

6.3.2 Login page

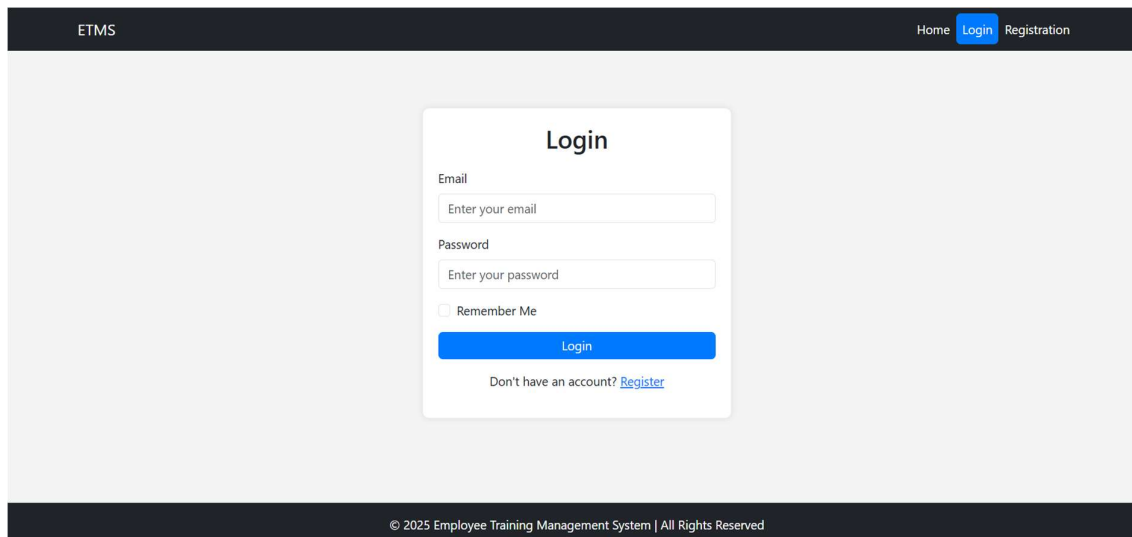


Fig 6.2 Login Page

6.3.3 Registration page

ETMS Home Login Registration

Register

Username
Enter your username

Email
Enter your email

Password
Create a password

Confirm Password
Confirm your password

Role
-- Select Role --

Register

Already have an account? [Login](#)

Fig 6.3 Registration Page

6.3.4 Trainer Dashboard

Trainer Dashboard Welcome, Trainer1

- Dashboard
- Manage Modules
- Manage Training
- Employee Progress
- Log Out

Manage Modules
Organize and update training modules.

Manage Training
Schedule and track training sessions.

Employee Progress
Monitor employee learning progress.

Fig 6.4 Trainer Dashboard

6.3.5 Manage Modules

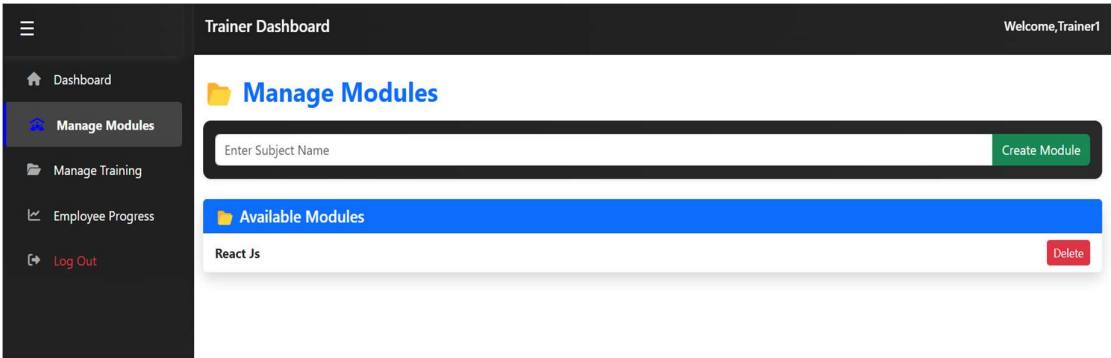


Fig 6.5 Manage Modules

6.3.6 Manage Training

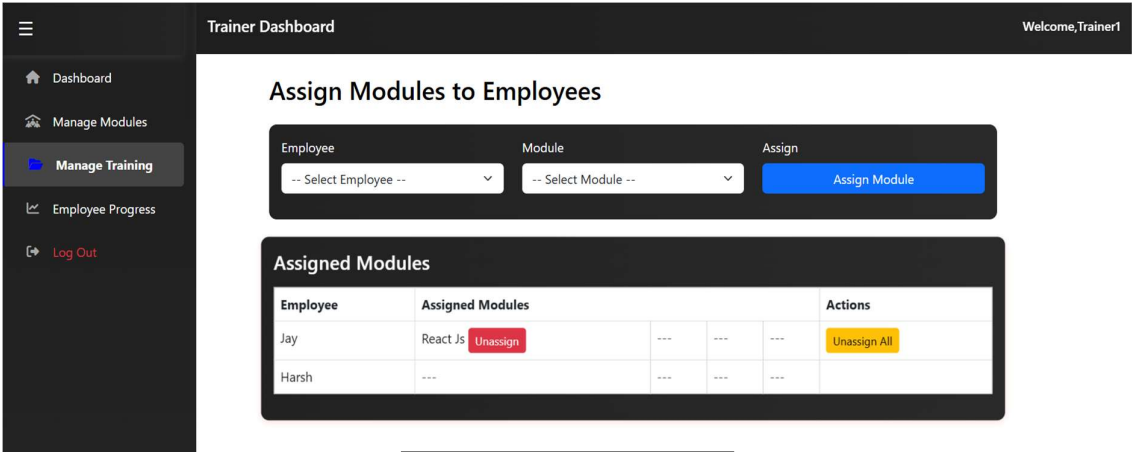


Fig 6.6 Manage Training

6.3.7 Manage Project

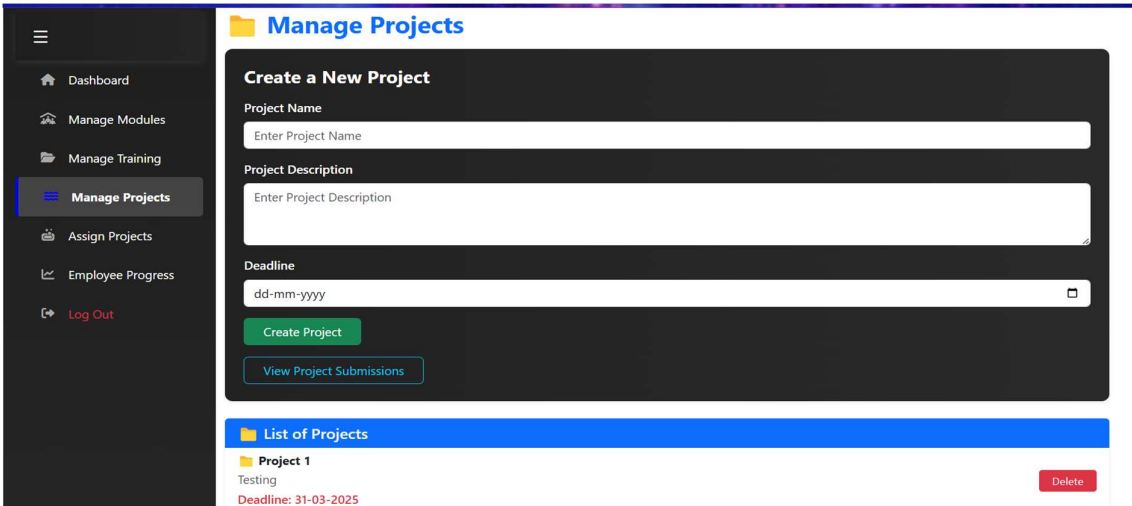


Fig 6.7 Manage Project

6.3.8 Assign Project

Trainer Dashboard Welcome, Trainer1

Assign Projects to Employees

Employee: -- Select Employee -- Project: -- Select Project -- Assign

Assign Project

Assigned Projects

Employee	Assigned Projects	Actions
Jay	Project 1 Unassign	Unassign All

Fig 6.8 Assign Project

6.3.9 Employees Progress Feedback

Trainer Dashboard Welcome, Trainer1

Employee Progress Feedback

Select Employee: -- Select Employee --

Test Marks (Auto-filled)

Select an employee to view test marks.

Project Marks

Enter marks out of 100

Feedback Evaluation

- Punctuality & Attendance**
 - Timeliness in attending training sessions
 - Adherence to break schedules
 - Consistency in attendance
- Participation & Engagement**
 - Level of interaction during training

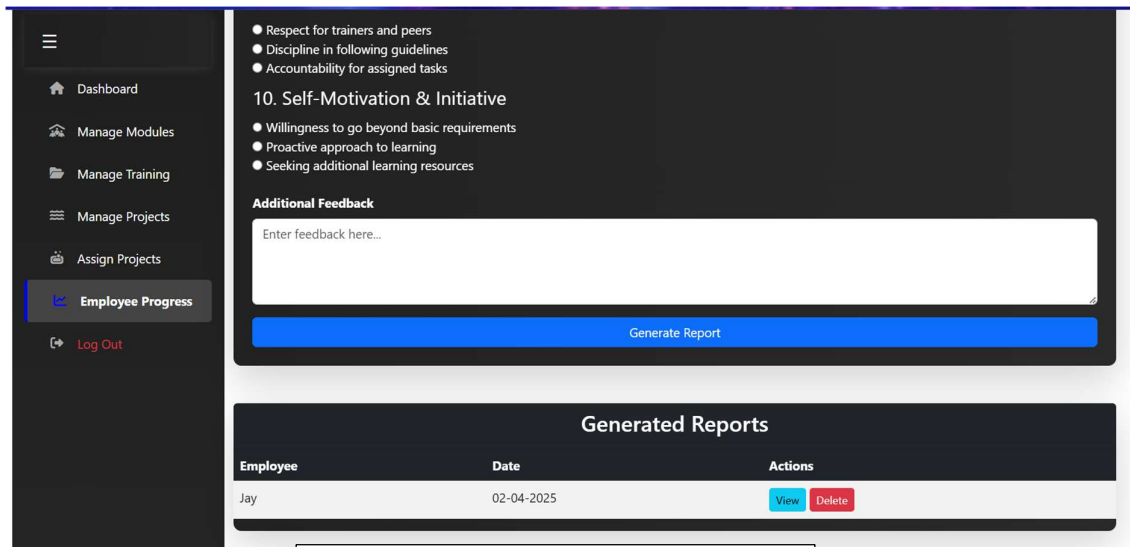


Fig 6.9 Employees Progress Feedback

6.3.10 Admin Dashboard

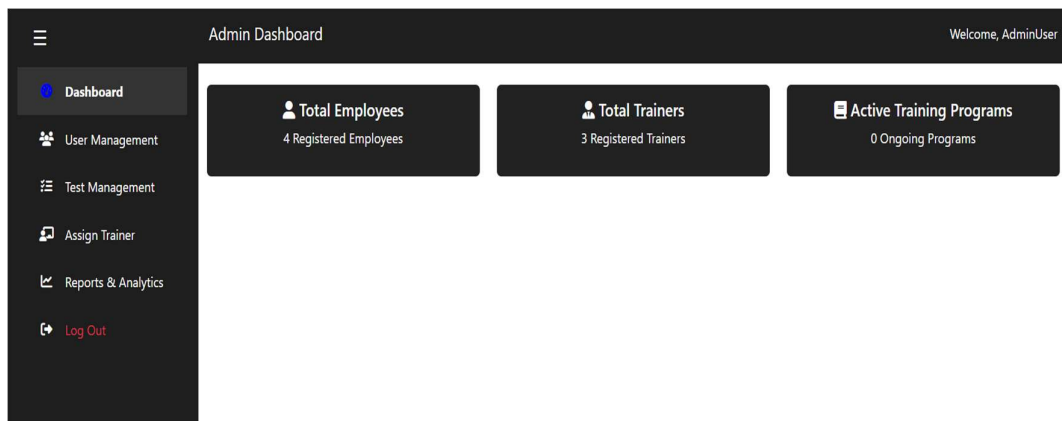


Fig 6.10 Admin Dashboard

6.3.11 User Management

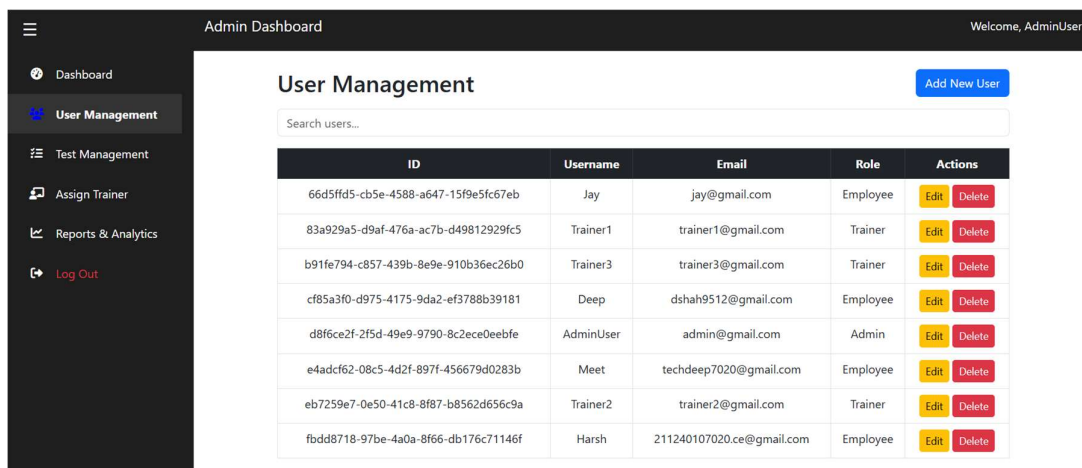


Fig 6.11 User Management

6.3.12 Test Management

Admin Dashboard Welcome, AdminUser

Test Management

Test Type	Mode	Actions
Pre-training	Active	Create View Test Status
Mid-training	Inactive	Create View Test Status
Post-training	Inactive	Create View Test Status

Notify Employees for Test

Select Test Type: Test Date: Test Time:

[Notify Employees](#)

Fig 6.12 Test Management

6.3.13 Assign trainer

Admin Dashboard Welcome, AdminUser

Assign Employees to Trainers

Trainer: Employee 0+1: Employee 1+1: Employee 2+1: Employee 3+1:

[Assign](#)

Assigned Employees

Trainer	Assigned Employees	Actions
Trainer1	Jay Unassign Harsh Unassign --- ---	Unassign All
Trainer3	Deep Unassign --- ---	Unassign All
Trainer2	Meet Unassign --- ---	Unassign All

Fig 6.13 Assign Trainer

6.3.14 Employee Dashboard

Employee Dashboard Welcome, Jay

My Tests
View upcoming and past tests.

My Training
Access training modules.

My Performance
Track your test results.

Fig 6.14 Employee Dashboard

6.3.15 My tests

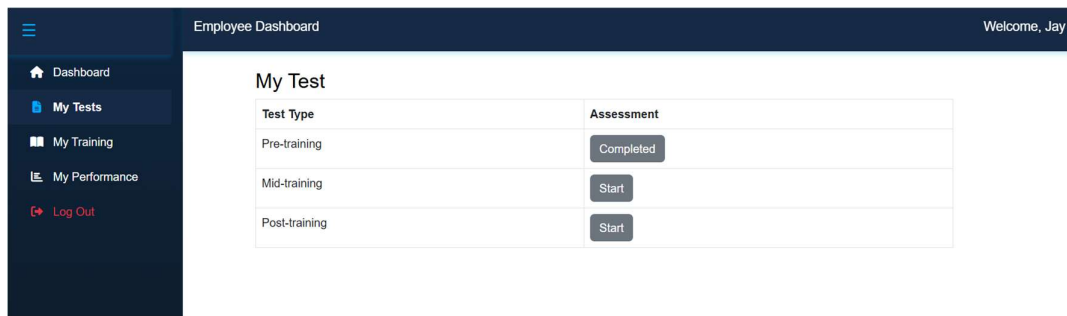


Fig 6.15 My Tests

5.3.16 My Training

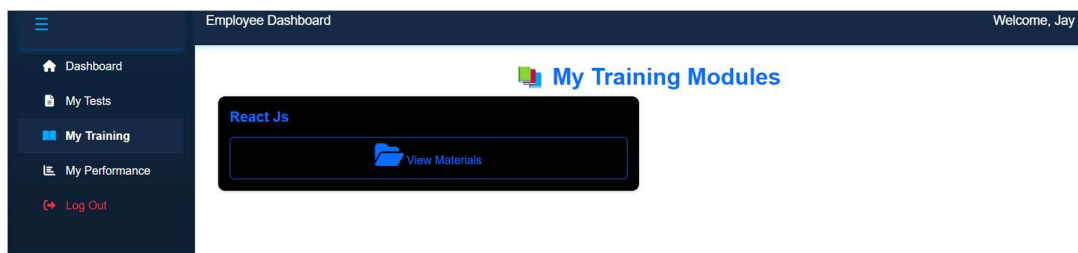


Fig 6.16 My Training

6.3.17 Test

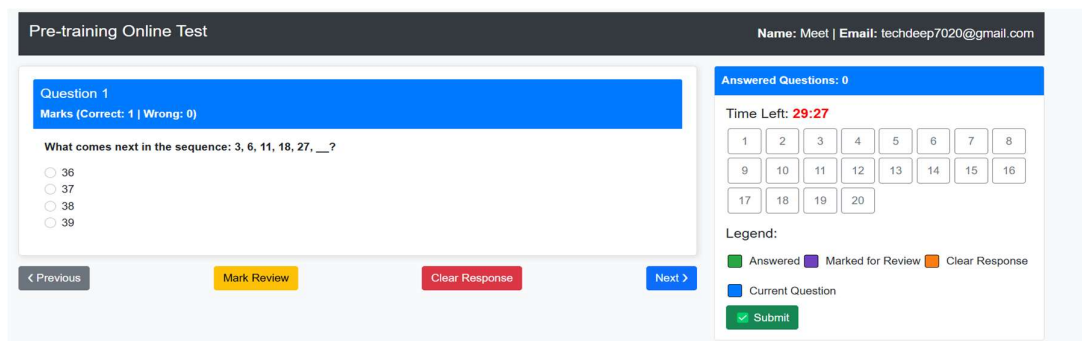


Fig 6.17 Test

6.3.18 Materials

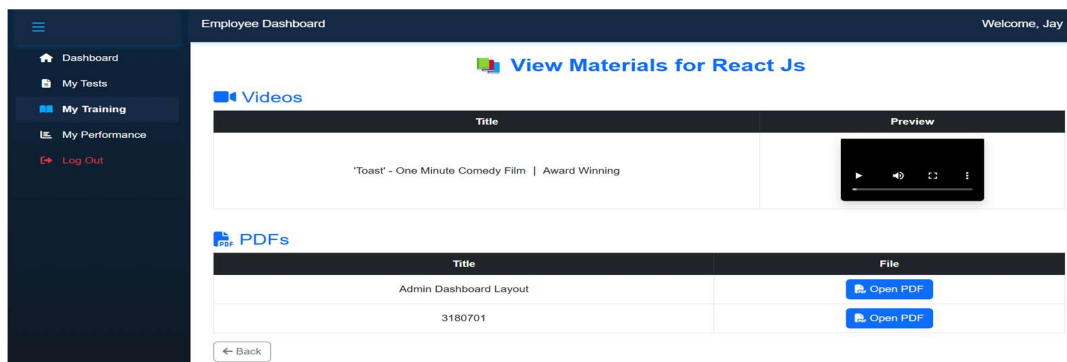


Fig 6.18 Materials

6.3.19 My Project

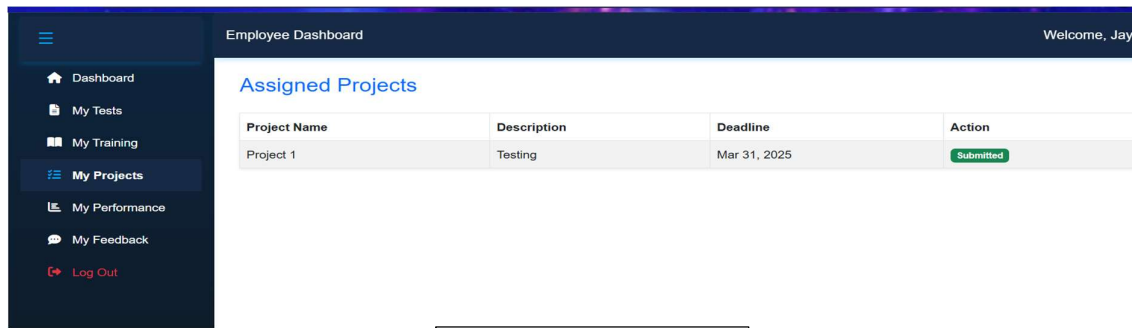


Fig 6.19 My Project

6.3.20 My Performance

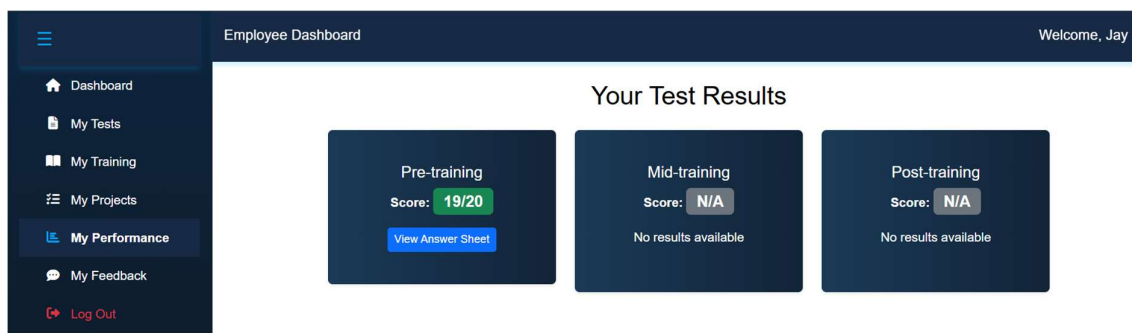


Fig 6.20 My Performance

6.3.21 My Feedback

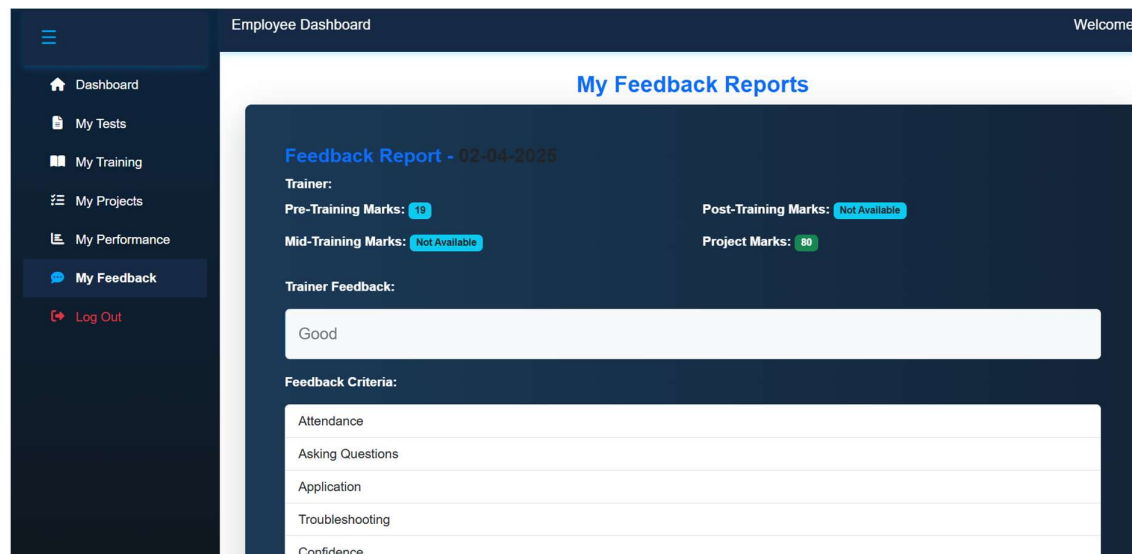


Fig 6.21 My Feedback

6.4 RESULT ANALYSIS:

The results of the Employee Training Management System (ETMS) implementation were analyzed and compared with traditional training management methods to evaluate its effectiveness and identify areas for improvement.

Efficiency Gains:

- Training Material Management:
 - The time required for trainers to upload and assign training materials was reduced from 30 minutes (manual process) to **5** minutes, an **83%** improvement.
 - Employees can access learning materials instantly, eliminating delays in material distribution that previously took 1-2 days.
- Assessment Process:
 - Online assessments reduced evaluation time from **3-5** days (manual grading) to instant results, improving efficiency by **90%**.
 - Trainers can review results immediately, speeding up feedback and employee performance tracking.
- Performance Evaluation:
 - Employee performance reports are now automatically generated, reducing report generation time from **2-3** hours to **5-10** minutes, an over **90% efficiency** gain.

Accuracy:

- Error Reduction in Training Assignment:
 - Previously, **10-15% of** employees reported incorrect training assignments. With ETMS, automated role-based material allocation reduced errors to **0%**.
- Assessment Accuracy:
 - Manual grading had an **estimated 5-10%** error rate, whereas the automated system ensures **100%** accuracy in test evaluations.

CHAPTER 7

7.0 TESTING

7.1 Testing Plan / Strategy

The testing strategy for the **Employee Training Management System (ETMS)** was designed to ensure that all modules function correctly and meet the specified requirements. The main objective was to detect any bugs or issues and ensure smooth performance across all functionalities. The testing was carried out in the following stages:

- **Unit Testing:** Each individual module/component (e.g., login, registration, test submission) was tested in isolation using test data.
- **Integration Testing:** After unit testing, modules were integrated, and their interactions were tested (e.g., test assignment and auto-submission logic).
- **System Testing:** The entire application was tested as a whole to ensure all features worked together seamlessly.
- **User Acceptance Testing (UAT):** The system was tested from the end-user perspective (admin, trainer, and employee) to validate that it fulfilled real-world expectations.

Tools Used:

- Manual testing
- Visual Studio Test Explorer
- SQL Server queries for backend data validation

7.2 Test Results and Analysis

After performing the above testing strategy, the system was found to be stable with all core functionalities working as expected. Minor UI bugs and logical errors were identified and resolved during the development phase. The table below shows sample test cases and their outcomes.

7.2.1 Test Cases

Test Case ID	Test Case Description	Input	Expected Output	Actual Output	Status
TC001	Employee Registration	Valid user details	Account created,	Success, user redirected to	Pass

			redirected to login	login	
TC002	Invalid Login Attempt	Wrong password/email	Error message displayed	Error: Invalid credentials	Pass
TC003	Assign Test to Employee	Select test and employee	Test assigned successfully	Assignment success message shown	Pass
TC004	Auto Submit on Test Timeout	Start test, exceed time limit	Test auto-submitted after timeout	Auto-submit triggered correctly	Pass
TC005	Prevent Re-attempt of Test	Try to re-open submitted test	Access denied message	Test already submitted message shown	Pass
TC006	Upload Learning Material (Trainer)	Select and upload PDF	Material uploaded and visible to assigned employees	File uploaded and listed	Pass
TC007	Project Submission (Employee)	Upload ZIP file	File uploaded and visible to trainer	ZIP successfully uploaded	Pass
TC008	Generate Performance Report (Trainer/Admin)	Select employee, enter marks, submit	Report generated with correct data	Report listed with marks and feedback	Pass

Chapter 8

8.0 CONCLUSION AND DISCUSSION

8.1 Overall Analysis of Internship

The internship at **Tech Elecon Pvt. Ltd.** provided valuable real-world exposure to software development practices and methodologies. Working on the **Employee Training Management System (ETMS)** project allowed me to understand and participate in every stage of the Software Development Life Cycle (SDLC), including requirement gathering, system design, coding, testing, and deployment. The experience enhanced my technical skills in **ASP.NET Core MVC**, **Entity Framework Core**, and **SQL Server**, as well as strengthened my soft skills like teamwork, time management, and communication.

8.2 Dates of Continuous Evaluation (CE-I and CE-II)

Evaluation	Date	Description
CE-I	15-02-2025	Evaluation of initial planning and design
CE-II	20-03-2025	Evaluation of development and testing

8.3 Problem Encountered and Possible Solutions

Problem	Possible Solution Implemented
Role-based access errors during login	Refined Identity role claims and user validation logic
Frontend alignment issues on different devices	Used responsive Bootstrap classes and media queries
Auto-submit of test not triggering properly	Added accurate Timer and session timeout control using JavaScript and C#
Duplicate project assignment	Added server-side validation to prevent duplicate entries
Difficulty in generating dynamic reports	Implemented backend LINQ queries to fetch marks and bind feedback dynamically

8.4 Summary of Internship

The internship experience was enriching both technically and professionally. Throughout the development of ETMS, I had the opportunity to:

- Develop real-time modules including test assignment, report generation, and project submission.
- Apply theoretical knowledge of databases and software architecture to a live project.
- Gain hands-on experience in ASP.NET Core, MVC, SQL Server, GitHub, and Visual Studio.
- Understand the importance of code optimization, UI/UX, testing, and deployment processes.

This internship served as a stepping stone in my journey toward becoming a full-stack developer.

8.5 Limitations and Future Enhancements

Limitations	Future Enhancements
Currently accessible on local network or IIS setup only	Hosting on cloud platforms (e.g., Azure, AWS) for wider accessibility
Limited reporting templates	Addition of downloadable PDF/Excel reports for performance analysis
Manual project review process	Include automated project evaluation tools or plagiarism checker
No mobile application	Development of a mobile-friendly version or dedicated Android/iOS app

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